



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)

FULL SYLLABUS

15-05-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-I)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	B	A	C	C	B	B	B	B	D
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	C	B	A	C	C	A	B	A	B
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	121	4	8	5	2	480	3	0	16	3

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	D	C	D	D	C	B	A	D	C
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	B	A	D	D	A	D	B	B	C
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	5	5	6	225	13	3	37	3	6	4

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	C	D	C	D	A	B	B	C	C
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	C	B	B	D	B	D	D	A	B
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	640	1	4	6	6	36	1	2	2	7

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1.

Ans (A)

$$\omega = \frac{2\pi}{t} = \frac{2\pi}{60} = \frac{\pi}{30} \text{ rad/min}$$

$$v = r\omega = 10 \times \frac{\pi}{30} = \frac{\pi}{3} \text{ cm/min.}$$

$$a_{\text{av}} = \frac{\Delta v}{\Delta t} = \frac{\sqrt{2}V}{15} = \frac{\sqrt{2} \times \pi}{3 \times 15} = \frac{\sqrt{2}\pi}{45} \text{ cm/min}^2$$

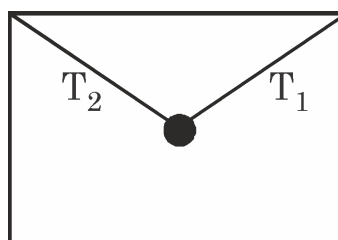
2.

Ans (B)

$$\vec{v}_{\text{WS}} = \vec{v}_{\text{WG}} + \vec{v}_{\text{GR}} + \vec{v}_{\text{RS}} = 6\hat{j} - 2\hat{j} + 3\hat{i} = 3\hat{i} + 4\hat{j}$$

3.

Ans (A)



$$T_1 \sin 30^\circ + T_2 \sin 30^\circ = mg$$

$$2mg \times \frac{1}{2} + T_2 \frac{1}{2} = mg$$

$$T_2 = 0$$

$$T_1 \cos 30^\circ = ma.$$

$$2mg \times \frac{\sqrt{3}}{2} = ma$$

$$a = \sqrt{3}g$$

4. **Ans (C)**

$$F_{\text{net}} = \sqrt{(mg)^2 + (ma_t)^2}$$

$$= m\sqrt{g^2 + (R\alpha)^2}$$

$$= m\sqrt{(10)^2 + (10)^2}$$

$$= 10 \times 10\sqrt{2}$$

$$= 100\sqrt{2}\text{N} = 141\text{N}$$

$$N_1 = m\omega^2 R = 10 \times 100 \times 2 = 2000\text{N}$$

$$F_{\text{max}} = \mu N = 0.2 \times 2000 = 400\text{N}$$

5. **Ans (C)**

Some mass of cork is out of water so

$$y_{\text{cm}} > \frac{H}{2}$$

$$x_{\text{cm}} = R.$$

6. **Ans (B)**

h will decrease as the density of metal is higher than water.

7. **Ans (B)**

The permeability is given by $\mu = B/H$

$\therefore$ Relative permeability

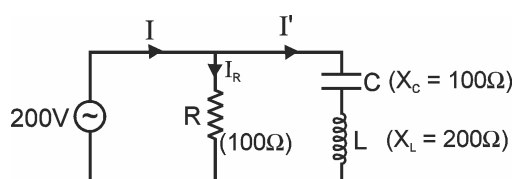
$$\mu_r = \frac{\mu}{\mu_0} = \frac{B}{\mu_0 H} = \frac{10}{4\pi \times 10^{-7} \times 250} = \frac{10^5}{\pi}$$

8. **Ans (B)**

Theory

9.

Ans (B)



$$I_R = 2$$

$$I' = \frac{V}{X_L - X_C} = \frac{200}{200 - 100} = 2$$

$$I_R = \frac{V}{R} = \frac{200}{100} = 2\text{A}$$

$$I' = \frac{V}{X_L - X_C} = \frac{200}{200 - 100} = 2\text{A}$$

$$I = \sqrt{I_R^2 + I'^2} = 2\sqrt{2}\text{Amp.}$$

10. **Ans (D)**

$$P = \frac{F}{A} = \frac{BIL}{A}$$

$$P = \frac{dF}{dA} = \frac{\mu_0(I_0 - \frac{1}{2}I)}{2\pi R} \frac{I}{2\pi R} (Rd\theta) \times \frac{L}{(LRd\theta)} = \frac{\mu_0(2I_0 - I)I}{8\pi^2 R^2}$$

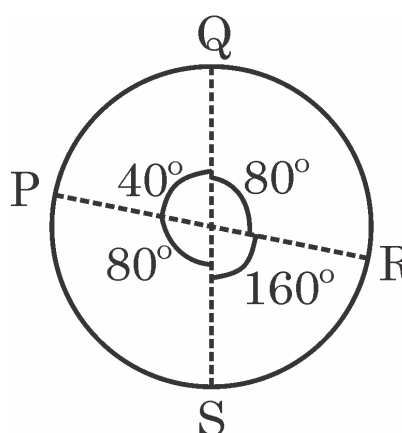
11. **Ans (C)**

Theory

12. **Ans (C)**

Refer notes

13. **Ans (B)**



$$\frac{R_{PQ}}{R_{QR}} = \frac{R_{PS}}{R_{SR}}$$

$$\text{So } \rightarrow V_P = V_R$$

$$\text{So current} = 0$$

14. **Ans (A)**

$$T \propto \sqrt{I}$$

15. **Ans (C)**

Ray will emerge out of the liquid up to critical angle.

$$\text{Fraction escaping} = \frac{2\pi(1 - \cos \theta)}{4\pi} = \frac{1 - \cos \theta}{2}$$

$\theta \rightarrow$ critical angle which does not depend on a depth.

16. **Ans (C)**

$$f_0 + f_e = 220$$

$$\frac{f_0}{f_e} = 10$$

$$f_e = 20$$

$$f_0 = 200$$

$$\text{Now } \rightarrow v = 100, f_e = 20$$

$$-\frac{1}{100} - \frac{1}{u} = \frac{1}{20}$$

$$\frac{1}{u} = -\frac{1}{100} - \frac{1}{20}$$

$$u = \frac{-100}{6}$$

$$\Delta H = 20 - \frac{100}{6}$$

$$\Delta H = \frac{20}{6} = \frac{10}{3} \text{ cm}$$

17. **Ans (A)**

From screw gauge

zero error = + 0.20 mm

Dia of Chalk = 3.5 - 0.2

= 3.3 mm

for vernier calliper

zero error = - 0.05 mm

Thus it must show = 0.33 - 0.05

= 0.28 mm

18. **Ans (B)**

$$\frac{dN}{dt} = \lambda N_0$$

$$t = \frac{60}{50} = 1.2 \text{ sec.}$$

19. **Ans (A)**

Energy incident of cobalt

$$= \frac{1200}{200} = 6 \text{ eV}$$

$$K_{\max} = 6.0 - 5.00 = 1.0 \text{ eV}$$

$$\frac{KE_{\max}}{KE_{\min}} = \frac{\frac{\sigma_e}{\epsilon_0} \times \ell + K_{\max}}{\frac{\sigma_e}{\epsilon_0} \ell}$$

$$= \frac{10 + 1.0}{10} = \frac{11.0}{10}$$

$$= 1.10$$

20. **Ans (B)**

S_2 is leading by $\frac{\pi}{4}$ in initial phase and S_2 also lead by $K\Delta x$ due to path difference

$$\phi_{\text{total}} = \frac{\pi}{4} + K\Delta x$$

$$= \frac{\pi}{4} + \frac{2\pi}{\lambda} \times \frac{\lambda}{8} = \frac{\pi}{2}$$

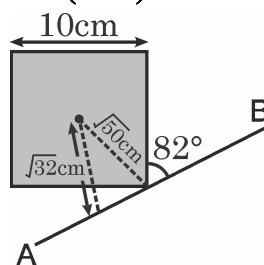
$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$$

$$= I_0 + I_0 + 2I_0 \cos\left(\frac{\pi}{2}\right) = 2I_0$$

PART-1 : PHYSICS

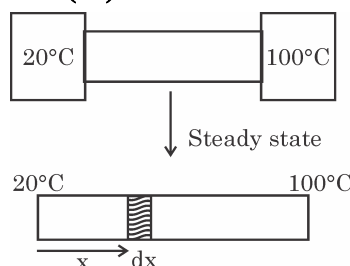
SECTION-II

1. **Ans (121)**



$$\begin{aligned} I_{AB} &= I_{\text{cm}} + Md^2 \\ &= \frac{ML^2}{12} + Md^2 \\ &= \frac{1 \times (10)^2}{12} + 1 \times 32 \\ &= \frac{25}{3} + 32 \\ &= \frac{121}{3} = 40.33 \text{ kg} - \text{cm}^2 \end{aligned}$$

2. **Ans (4)**



Temperature at any distance x is

$$T_x = (20 + 40x).$$

Change in temp. $\Rightarrow (20 + 40x) - 20 = 40x$

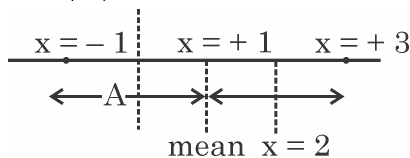
Heat absorbed $d\theta = dm \cdot \delta \cdot (40x)$

$$dm = \frac{2\text{kg}}{2\text{m}} \cdot dx = dx$$

$$\int d\theta = \delta \cdot \int_0^2 40x dx$$

$$= \delta \times 80 = 50 \times 80 = 4000 \text{ J.}$$

3. Ans (8)



$$\frac{T}{4} = 3 \text{ sec.}$$

$$T = 12 \text{ sec.}$$

$$t_x = 2(\text{second time})$$

$$= \frac{T}{4} + \frac{T}{4} + \frac{T}{6}$$

$$= \frac{8T}{12} = 8 \text{ sec.}$$

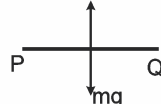
4. Ans (5)

The F.B.D. of wire PQ is

$$\text{The force due to surface tension} = F_{ST} = 2T \times 2$$

$$AD \tan \theta$$

$$F_{ST} = (2T) 2AD \tan \theta$$



$$F_{ST} = (2T) 2(AD + x) \tan \theta$$

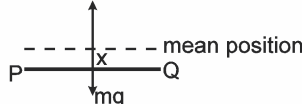


Figure (a)

Figure (b)

For wire to be in equilibrium (Figure (a)) $4T AD \tan \theta = mg$ (1)

If the wire PQ is at a distance x below the mean position, the restoring force on the wire is (Figure (b))

$$-ma = 4T \tan \theta (AD + x) - mg = 4T \tan \theta x$$

Hence the wire PQ executes SHM

$$a = -\frac{4T}{m} \tan \theta x$$

comparing with $a = -\omega^2 x$ we get

$$\omega^2 = \frac{4T}{m} \tan \theta$$

$$\text{or } T = 2\pi \sqrt{\frac{m}{4T \tan \theta}} = 2\pi \sqrt{\frac{1 \times 10^{-3}}{4 \times 25 \times 10^{-3}}} \\ = \frac{\pi}{5} \text{ s}$$

5. Ans (2)

Refer notes

6. Ans (480)

$$E = 4 \times 5 \times 1 = 20$$

$$W_{\text{Ext}} = \frac{(20)^2}{4} \times 3 + \frac{1}{2} \times 10 \times (6)^2$$

$$= 300 + 180 \Rightarrow 480$$

7. Ans (3)

$$W_E = 2 \times \int_{(3,3,1)}^{(4,2,1)} x dx + 2y dy + 3dz = -3$$

8. Ans (0)

$$\rho = \lambda_0 (3 - x)^2$$

$$q_{\text{in}} = \int_0^x \lambda_0 (3 - x)^2 4\pi x^2 dx$$

$$= 4\pi \lambda_0 \int_0^x (3^2 x^2 + x^4 - 2.3x^3) dx$$

$$q_{\text{in}} = 4\pi \lambda_0 \left(\frac{3^2 x^3}{3} + \frac{x^5}{5} - \frac{3x^4}{2} \right)$$

$$E = \frac{\lambda_0}{E_0} \left(\frac{a^2 x}{3} + \frac{x^3}{5} - \frac{ax^2}{2} \right)$$

$$\frac{a^2}{3} + \frac{3x^2}{5} - ax = 0$$

$$3 + \frac{3x^2}{5} - 3x = 0$$

$$5 + x^2 - 5x = 0$$

$$x^2 - 5x + 5 = 0$$

$$\frac{5 \pm \sqrt{25 - 20}}{2}$$

$$x = \frac{5 \pm \sqrt{5}}{2}$$

9. Ans (16)

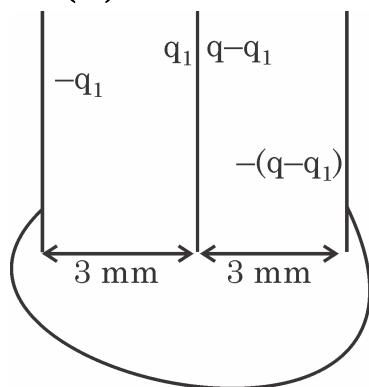
$$-\frac{GMm}{R} + \frac{1}{2} m \left(\frac{9}{16} \times \frac{GM}{R} \right) = -\frac{GMm}{(R+h)}$$

$$-\frac{7}{16} \frac{GMm}{R} = -\frac{GMm}{(R+h)}$$

$$R+h = \frac{16R}{7}$$

$$h = \frac{9R}{7}$$

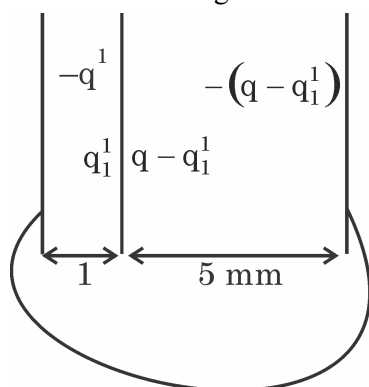
10. Ans (3)



$$\frac{q_1}{A\epsilon_0} 3 = \frac{q - q_1}{A\epsilon_0} 3$$

$$q_1 = \frac{q}{2}$$

Now after shifting



$$\frac{q_1^1}{A\epsilon_0} 1 = \frac{q - q_1^1}{A\epsilon_0} 5$$

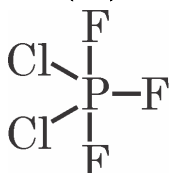
$$q_1^1 = \frac{5q}{6}$$

$$\Rightarrow \Delta q = \frac{q}{3} = \frac{9}{3} = 3\mu C$$

PART-2 : CHEMISTRY

SECTION-I

1. Ans (D)



Axial and equatorial bond lengths of P-F bonds are not same.

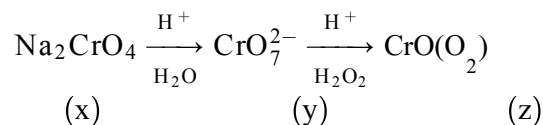
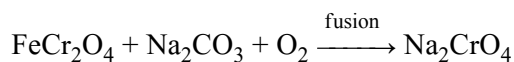
2. Ans (D)

Gibbs free energy of formation of most sulphides are less than that for CS₂.

3. Ans (C)

Fact.

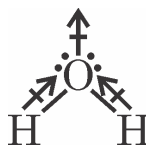
4. Ans (D)



Deep blue colour solution

CrO(O₂)₂ decomposes rapidly in aqueous solution into Cr³⁺ and oxygen saturated solution dichromate ion gives bright orange compound with conc. H₂SO₄.

5. Ans (D)

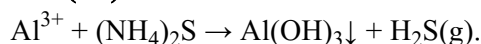


In H₂O overall dipole moment add up so it is slightly higher.

6. Ans (C)

[Pt(NH₃)₄] [PtCl₆] can interchange ligands so it can show coordination isomerism.

7. Ans (B)



8. Ans (A)

$$\text{Ionic mobility, } \mu^0 = \frac{\text{Speed of ion}}{\text{Potential gradient}} = \frac{\lambda_m^0}{F}$$

$$\frac{\text{Speed}}{\left(\frac{38.6}{6}\right)} = \frac{50}{96500} = 3.33 \times 10^{-3}$$

9. Ans (D)



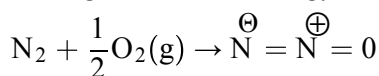
$$\frac{31.82 - 31.78}{31.82} = \frac{3S}{55.5}$$

$$S = 2.33 \times 10^{-2}$$

$$K_{sp} = 4s^3 = 5.06 \times 10^{-5}$$

10. **Ans (C)**

We calculate theoretical value of $\Delta_f H^\circ$ of N_2O .
Difference of experimental and theoretical value give resonance energy



$$\Delta_f H^\circ(N_2O) = [(B.E)_{N \equiv N} + \frac{1}{2}(BE)_{O=O}]$$

$$- [(BE)_{N=N} + (BE)_{N=O}]$$

$$= \left[950 + \frac{500}{2} \right] - (420 + 610) = 170$$

$$\Delta_f H^\circ (\text{experimental}) = 82 \text{ KJ}$$

$$\text{Resonance energy } N_2O = 82 - 170$$

$$= -88 \text{ KJ/mol}$$

11. **Ans (D)**

When phenolphthalein is used as indicator

$$V.f \text{ of } Na_2CO_3 = 1$$

$$\text{and } V.f \text{ of } NaOH = 1$$

$$\text{gm equiv.} = \text{gm equiv} + \text{gm equiv.}$$

$$\text{of HCl} \quad \text{of } Na_2CO_3 \quad \text{of NaOH}$$

$$\frac{1 \times V(\text{ml})}{1000} = \left(\frac{2}{106} \times 1 \right) + \left(\frac{2}{40} \right)$$

$$V = 68.86 \text{ ml}$$

12. **Ans (B)**

Refer notes.

13. **Ans (A)**

$$K = Ae^{-E/RT} \quad \dots(1)$$

$$10^6 k = Ae^{-E_c/RT} \quad \dots(2)$$

$$\frac{\text{equation 2}}{\text{equation 1}} = 10^6 = e^{(E-E_c)/RT}$$

$$6 \ln 10 = (E - E_c)/RT$$

$$\frac{(E - E_c)}{RT} = 2.303 \times 6$$

$$E_c - E = -2.303 \times 6RT$$

14. **Ans (D)**

Refer notes.

15. **Ans (D)**

Refer notes.

16. **Ans (A)**

Fact

17. **Ans (D)**

Hoffman bromamide degradation reaction.

18. **Ans (B)**

Fact

19. **Ans (B)**

Fact

20. **Ans (C)**

Refer notes.

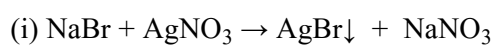
PART-2 : CHEMISTRY

SECTION-II

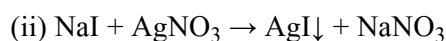
1. **Ans (5)**

Refer notes.

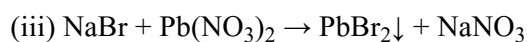
2. **Ans (5)**



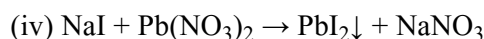
light yellow



yellow

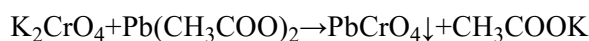


white

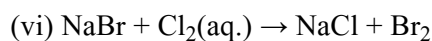


yellow

(v)



yellow



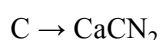
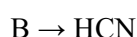
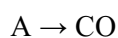
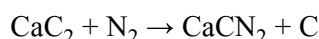
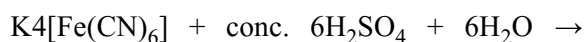
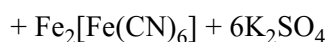
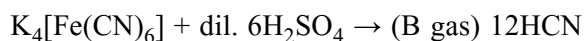
yellow

5 ppt giving yellow colour.

3. **Ans (6)**



(excess) (x)



Total p bond = 6.

4. **Ans (225)**

$$2 = \frac{Z \times (75/6 \times 10^{23})}{(5 \times 10^{-8})^3}$$

$$Z = 2$$

$$4r = a\sqrt{3}$$

$$r = \frac{5 \times 1.8}{4} = 2.25 \text{ Å or } 225 \text{ pm}$$

5. **Ans (13)**



$$K_a = 5.55 \times 10^{-10}$$

$$\text{so } (K_b)_{\text{NH}_3} = \frac{K_w}{K_a} = \frac{10^{-14}}{5.55 \times 10^{-10}} = 1.8 \times 10^{-5}$$

For basic buffer

$$\text{pOH} = \text{p}K_b + \log \frac{[\text{NH}_4^+]}{[\text{NH}_3]}$$

$$\text{pOH} = \text{p}K_b \Rightarrow (\text{OH}^-) = K_b = 1.8 \times 10^{-5}$$

$$[\text{NH}_4^+] = [\text{NH}_3]$$

$$\text{New } K_{sp} = [\text{Mn}^{2+}] [\text{OH}^-]^2$$

$$[\text{Mn}^{2+}] = \frac{K_{sp}}{[\text{OH}^-]^2} = \frac{4.212 \times 10^{-14}}{(1.8 \times 10^{-5})^2} = 1.30 \times 10^{-4}$$

$$x = 13.8$$

6. **Ans (3)**

Refer notes

7. **Ans (37)**

$$5.58 = \frac{1.86 \times 11.718 \times 1000}{62 \times W}$$

$$W = \frac{11.718 \times 1000 \times 1.86}{5.58 \times 62} = 63$$

$$\text{Ice separated} = 100 - 63 = 37 \text{ g}$$

8. **Ans (3)**

Refer notes

9. **Ans (6)**

Refer notes

10. **Ans (4)**

Refer notes

PART-3 : MATHEMATICS

SECTION-I

1. **Ans (B)**

$$\operatorname{cosec} \frac{\pi}{4} = \frac{1}{2} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 0 & 0 & (\lambda - 1) \end{vmatrix}$$

$$2\sqrt{2} = |(\lambda - 1)\hat{i} - (\lambda - 1)\hat{j}|$$

$$2\sqrt{2} = |\lambda - 1|\sqrt{2}$$

$$\lambda = 3, -1$$

2. **Ans (C)**

$$f(5) = 0 \Rightarrow 50 + 5a + 5 = 0$$

$$a = -11$$

$$2x^2 - 11x + 5 = 0$$

$$x = 5, \frac{1}{2}$$

$$(L-H \text{ Rule}) \lim_{x \rightarrow \frac{1}{2}} \frac{e^{f(x)} - 1}{2x - 1} = \frac{e^{f(x)} \cdot f'(x)}{2}$$

$$= \frac{e^{f(\frac{1}{2})} \cdot f'(\frac{1}{2})}{2}$$

$$= \frac{-9}{2}$$

3. **Ans (D)**

$$5[x] + 5\{x\} + 2\{x\} = 3[x] + 4$$

$$7\{x\} = 4 - 2[x]$$

$$0 \leq 7\{x\} < 7$$

$$0 \leq 4 - 2[x] < 7$$

$$-3 < 2[x] \leq 4$$

$$\frac{-3}{2} < [x] \leq 2$$

$$[x] = -1, 0, 1, 2$$

$$\{x\} = \frac{6}{7}, \frac{4}{7}, \frac{2}{7}, 0$$

$$x = \frac{-1}{7}, \frac{4}{7}, \frac{9}{7}, 2$$

4. Ans (C)

Put $x = 1$

$$3^8 = a_0 + a_1 + a_2 + \dots + a_{16} \quad (1)$$

Put $x = -1$

$$1 = a_0 - a_1 + a_2 - \dots + a_{16} \quad (2)$$

Put $x = i$

$$i^8 = a_0 + a_1 i - a_2 + a_3 i + a_4 \quad (3)$$

$$(-i)^8 = a_0 - a_1 i - a_2 + a_3 i + a_4 \quad (4)$$

$$(1) + (2) + (3) + (4)$$

$$3^8 + 3 = 4[a_0 + a_4 + a_8 + a_{12} + a_{16}]$$

5. Ans (D)

$$\text{Let } \frac{a}{1+a} = x, \frac{b}{1+b} = y, \frac{c}{1+c} = z$$

$$x + y + z = 1, a = \frac{x}{1-x}, b = \frac{y}{1-y}, c = \frac{z}{1-z}$$

$$a = \frac{x}{y+z}, b = \frac{y}{z+x}, c = \frac{z}{x+y}$$

$$\frac{x+y}{2} \geq \sqrt{xy}, \frac{y+z}{2} \geq \sqrt{yz}, \frac{z+x}{2} \geq \sqrt{zx}$$

$$\frac{(x+y)(y+z)(z+x)}{xyz} \geq 8$$

$$abc \leq \frac{1}{8}$$

6. Ans (A)

$$L; \frac{x-1}{\frac{1}{2\sqrt{2}}} = \frac{y-2}{1} = \frac{z-0}{0}$$

$$\text{Plane}; \sqrt{2}x + y - z = 1$$

Angle between line and normal vector of plane

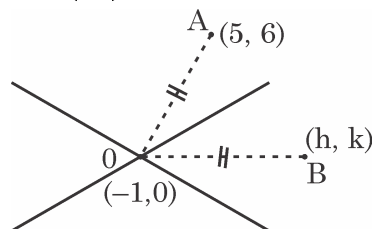
$$\cos \theta = \frac{\frac{1}{2\sqrt{2}} \times \sqrt{2} + 1}{\frac{3}{2\sqrt{2}} \times 2} = \frac{\frac{3}{2}}{\frac{3}{\sqrt{2}}} = \frac{1}{\sqrt{2}}$$

$$\theta = 45^\circ$$

So distance is measured perpendicular to the plane.

$$PQ = \left| \frac{\sqrt{2} + 0 - 1 + 1}{2} \right| = \frac{1}{\sqrt{2}}$$

7. Ans (B)



$$(2x + y + 2) - m(x - y + 1) = 0$$

$$2x + y + 2 = 0$$

$$x - y + 1 = 0$$

$$3x + 3 = 0$$

$$x = -1$$

$$y = 0$$

$$OA^2 = OB^2$$

$$(h+1)^2 + (k+0)^2 = 36 + 36$$

$$x^2 + y^2 + 2x - 71 = 0$$

8. Ans (B)

$$x > 0$$

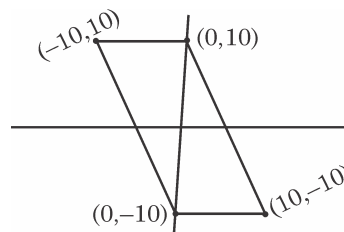
$$y^2 + 2xy + 20x = 100$$

$$y^2 - 100 + 2x(y + 10) = 0$$

$$(y + 10)(2x + y - 10) = 0$$

$$x < 0$$

$$(y - 10)(2x + y + 10) = 0$$



$$\text{Area} = 10 \times 20 = 200$$

9. Ans (C)

Let $P(h, k)$ be point of intersection

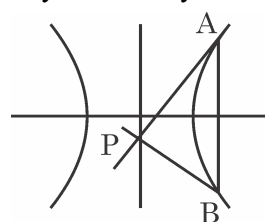
$$AB : xh - yk = 4$$

$$AB : \frac{x}{\sec \theta} + \frac{y}{\tan \theta} = 4$$

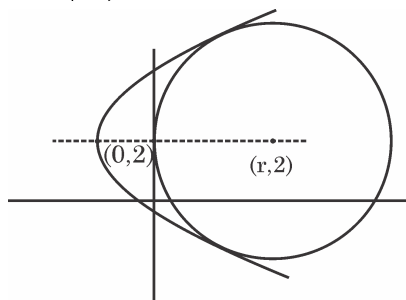
$$h \sec \theta = -k \tan \theta = 1$$

$$\frac{1}{h^2} - \frac{1}{k^2} = 1$$

$$y^2 - x^2 = x^2 y^2$$



10. Ans (C)



$$(y-2)^2 = 4(x+1)$$

Focus (0, 2)

$$(x-r)^2 + (y-2)^2 = r^2$$

$$(x-r)^2 + 4(x+1) = r^2$$

$$x^2 + (4-2r)x + 4 = 0$$

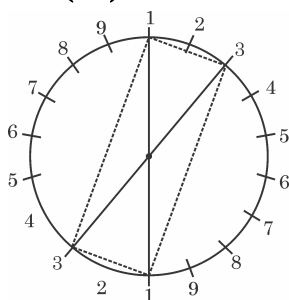
$$D = 0$$

$$4(2-r)^2 - 16 = 0$$

$$2-r = 2, -2$$

$$r = 4, 0$$

11. Ans (D)



No. of rectangles = ${}^9C_2 = 36$

12. Ans (C)

$$\int \frac{\sec^2 x}{1 + \sin^2 x} dx = \int \frac{(1 + \tan^2 x) \cdot \sec^2 x}{1 + \tan^2 x + \tan^2 x} dx$$

$$t = \tan x \quad \int \frac{1+t^2}{1+2t^2} dt$$

$$= \frac{1}{2} \int \frac{1+(1+2t^2)}{1+2t^2} dt$$

$$\frac{1}{2}t + \frac{1}{4} \int \frac{dt}{\frac{1}{2} + t^2}$$

$$\frac{1}{2}t + \frac{1}{4} \times \sqrt{2} \tan^{-1} \sqrt{2}t$$

$$\frac{1}{2} \tan x + \frac{1}{2\sqrt{2}} \tan^{-1} (\sqrt{2} \tan x)$$

$$\frac{1}{2} \tan x + \frac{1}{2\sqrt{2}} \tan^{-1} (\sqrt{2} \tan x) + D$$

$$a = \frac{1}{2}, \quad b = \frac{1}{2\sqrt{2}}, \quad c = \sqrt{2}$$

13. Ans (B)

$$S = 6(5 + \omega)(5 + \omega^2) + 7(6 + \omega)(6 + \omega^2) + \dots$$

$$S = \sum_{n=5}^{20} (n+1)(n+\omega)(n+\omega^2)$$

$$S = \sum_{n=5}^{20} (n+1)(n^2 - n + 1)$$

$$S = \sum_{n=5}^{20} (n^3 + 1)$$

$$S = \left(\left(\frac{20 \times 21}{2} \right)^2 - \left(\frac{4 \times 5}{2} \right)^2 \right) + 16$$

$$S = (210^2 - 10^2) + 16 \Rightarrow 220 \times 200 + 16$$

$$\Rightarrow 44016$$

14. Ans (B)

$$L = 2 \lim_{m \rightarrow \infty} \int_0^{\infty} \frac{dx}{1 + x^2 + x^4 + \dots + x^{2m}}$$

$$L = 2 \lim_{m \rightarrow \infty} \left[\int_0^1 \frac{dx}{1 + x^2 + x^4 + \dots + x^{2m}} + \int_1^{\infty} \frac{dx}{1 + x^2 + \dots + x^{2m}} \right]$$

$$= 2 \int_0^1 \frac{dx}{\frac{1}{1-x^2}} + 2(0)$$

$$= 2 \int_0^1 (1 - x^2) dx$$

$$= 2 \left(x - \frac{x^3}{3} \right)_0^1$$

$$= \frac{4}{3}$$

15. Ans (D)

$$(1, 2, 3, 4, 5) - (1, 2, 3, 4)$$

$$P = \frac{5^3 - 4^3}{6^3}$$

$$P = \frac{61}{216}$$

16. Ans (B)

$$|A| = a^2$$

$$\det(\text{adj}(\text{adj}(A))) = |A|^4 = a^8$$

$$a^8 = 16$$

$$a = \sqrt{2}$$

17. **Ans (D)**

$$p \Rightarrow (\sim p \vee q)$$

$$\sim p \vee (\sim p \vee q)$$

$$= (\sim p \vee q)$$

18. **Ans (D)**

$$f(x) = |\ln x| \quad g(x) = \lambda x$$

$f(x)$ and $g(x)$ touches at (x_1, y_1) in Ist Quadrant

$$\ln x_1 = y_1 \quad \dots (1)$$

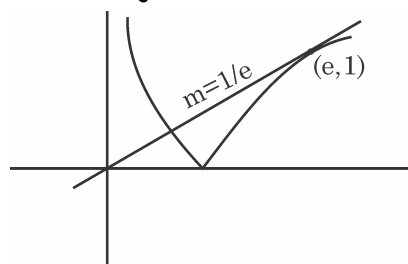
$$y_1 = \lambda x_1 \quad \dots (2)$$

$$f'(x_1) = g'(x_1) \quad \frac{1}{x_1} = \lambda \quad \dots (3)$$

By (1), (2), (3)

$$x_1 = e, y_1 = 1, \lambda = \frac{1}{e}$$

So for $\lambda > \frac{1}{e}$, line intersect the curve at one point.



19. **Ans (A)**

$$2x^3 dy + 5y^4 dy = 6x^2 y dx$$

$$5y^4 dy = 6x^2 y dx - 2x^3 dx$$

$$5y^2 dy = 2 d \left(\frac{x^3}{y} \right)$$

$$\frac{5y^3}{3} = 2x^3 \cdot y^{-1} + C$$

20. **Ans (B)**

For satisfying given condition

$$f'(x) = 3$$

$$f(x) = 3x + c$$

$$f(2) = 6 + c = 4 \Rightarrow c = -2$$

$$f(x) = 3x - 2$$

So $f(3) = 7$

$$f(4) = 10$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (640)**

$$\text{Tr}(A) = 0 \rightarrow (0, 0, 0), (-1, 0, 1), (1, 1, -2)$$

$$\text{Symmetric matrix} \rightarrow \begin{bmatrix} - & 4 & 4 \\ 1 & - & 4 \\ 1 & 1 & - \end{bmatrix}$$

$$4 \times 4 \times 4 = 64$$

$$\text{Total} = 64 \times 1 + 64 \times 3! + 64 \times 3$$

$$= 64 \times 10$$

$$= 640$$

2. **Ans (1)**

$$\text{Take } \frac{\ln x}{x} = t$$

$$\text{For } x \rightarrow \infty, t \rightarrow 0$$

$$\lim_{t \rightarrow 0} \frac{(3^t - 3^{\frac{t}{1+t}})}{t^2}$$

$$\frac{\left(1 + t \ln 3 + \frac{(t \ln 3)^2}{2!}\right) - \left(1 + \frac{t}{1+t} \ln 3 + \frac{\left(\frac{t}{1+t} \ln 3\right)^2}{2!}\right)}{t^2}$$

$$\lim_{t \rightarrow 0} \frac{t \ln 3 \left(\frac{t}{1+t}\right) + \frac{t^2 (\ln 3)^2}{2!} \left(1 - \frac{1}{(1+t)^2}\right)}{t^2}$$

$$\lim_{t \rightarrow 0} \frac{\ln 3}{1+t} + \frac{(\ln 3)^2}{2!} \left(\frac{t^2 + 2t}{(1+t)^2}\right)$$

$$\ln 3$$

3. **Ans (4)**

$$f(x) = x \ln x + \int_0^x e^{-(x-t)} f(t) dt$$

$$f'(x) = \ln x + 1 + e^{-x} \cdot e^x f(x) - e^{-x} \int_0^x e^t f(t) dt$$

$$f'(x) = \ln x + 1 + f(x) - [f(x) - x \ln x]$$

$$f'(x) = \ln x + 1 + x \ln x$$

$$f(x) = x \ln x + \int x \ln x \cdot dx$$

$$f(x) = x \ln x + (\ln x) \cdot \frac{x^2}{2} - \frac{x^2}{4} + C$$

$$f(1) = -\frac{1}{4} \Rightarrow C = 0$$

$$f(e) = e + \frac{e^2}{4}$$

4. **Ans (6)**

Total sample space = 30

Favourable case = 18

$$P = \frac{18}{30} = \frac{3}{5} = 0.6$$

5. **Ans (6)**

Take $z = (\cos\theta + i \sin\theta)$

$$\sqrt{|z|^2 - 6|z|\cos\theta + 9} + |z| + \sqrt{|z|^2 + 6|z|\cos\theta + 9}$$

$$= 12$$

$$|z|_{\max} \text{ when } \cos\theta = 1$$

$$|z| - 3 + |z| + 3 = 12$$

$$|z| = 6$$

6. **Ans (36)**

On shifting $|x| + |y| + |z| = 3$

Figure obtained is double pyramid

$$V = 2 \times \frac{1}{3} \times \text{area of base} \times \text{height}$$

$$V = 2 \times \frac{1}{3} \times 18 \times 3 = 36$$

7. **Ans (1)**

$$\Delta = \begin{vmatrix} \alpha & 1 & 1 \\ 1 & \alpha & 1 \\ 1 & 1 & \alpha \end{vmatrix} = 0$$

$$\alpha(\alpha^2 - 1) - 1(\alpha - 1) + 1(1 - \alpha) = 0$$

$$(\alpha - 1)(\alpha + 1 - 2) = 0$$

$$(\alpha - 1)2(\alpha + 2) = 0$$

$$\alpha = 1, \alpha = -2$$

✗ ✓

8. **Ans (2)**

Put $x = 2 \sin^2\theta$

$$I = \int_0^{\frac{\pi}{4}} \frac{4 \sin\theta \cos\theta}{\sqrt{2} \sin\theta + \sqrt{2} \cos\theta} d\theta$$

$$= \sqrt{2} \int_0^{\frac{\pi}{4}} \frac{\sin 2\theta}{\sin\theta + \cos\theta} d\theta$$

$$= \sqrt{2} \int_0^{\frac{\pi}{4}} \frac{\sin 2\theta + 1}{\sin\theta + \cos\theta} d\theta - \sqrt{2} \int_0^{\frac{\pi}{4}} \frac{1}{\sin\theta + \cos\theta} d\theta$$

$$= \sqrt{2} \left(\sin\theta - \cos\theta \right) \Big|_0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} \frac{d\theta}{\cos\left(\frac{\pi}{4} - \theta\right)}$$

$$= \sqrt{2} [1] - \int_0^{\frac{\pi}{4}} \frac{d\theta}{\cos\theta}$$

$$= \sqrt{2} \ln(\sec\theta + \tan\theta) \Big|_0^{\frac{\pi}{4}}$$

$$\sqrt{2} - \log(\sqrt{2} + 1)$$

9. **Ans (2)**

$$2x^2 + 11x + 8 = 0 \begin{matrix} \alpha \\ \beta \end{matrix}$$

$$\alpha + \beta = \frac{-11}{2}, \quad \alpha\beta = 4$$

$$S: x^2 + y^2 + 2gx + 2fy + C = 0$$

$$(\alpha, 0) \text{ and } (\beta, 0)$$

$$x^2 + 2gx + C = 0 \begin{matrix} \alpha \\ \beta \end{matrix}$$

$$\alpha + \beta = -2g = \frac{-11}{2} \quad \alpha\beta = C = 4$$

$$L_T = \sqrt{C} = 2$$

10. **Ans (7)**

$$\mu = 5 \quad \sigma^2 = 7$$

$$1 + 3 + 4 + 6 + x + y = 30$$

$$x + y = 16 \quad \dots (1)$$

$$7 = \frac{1}{6} \sum x_i^2 - \mu^2$$

$$7 = \frac{1}{6} [62 + x^2 + y^2] - 25$$

$$32 \times 6 - 62 = x^2 + y^2$$

$$x^2 + y^2 = 130 \quad \dots (2)$$

$$x = 9, y = 7$$

$$\text{Mean deviation} = \frac{\sum |x_i - \mu|}{6}$$

$$= \frac{4 + 2 + 1 + 1 + 4 + 2}{6}$$

$$= \frac{14}{6}$$

JEE(Main + Advanced) : ENTHUSIAST COURSE : [SCORE-I (ALL STAR BATCH)]

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	A,B,C,D	A,C	A,B,D	A,D	A,B	A,B	A,C,D	A,B,C
SECTION-I (ii)	Q.	9	10	11	12				
	A.	C	B	B	D				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	0.41	111.00 to 111.90	2.00	2.50 to 2.59	242.00 to 242.90	0.71 to 0.73		

PART-2 : CHEMISTRY

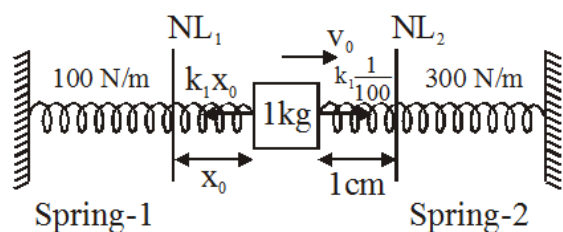
SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	A,B,D	A,C,D	B,C,D	B,C,D	A,C,D	A,C	A,C	A,B
SECTION-I (ii)	Q.	9	10	11	12				
	A.	C	D	A	D				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	3.00	109.00	33.33	2.25	4.00	2.50		

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	A,B,C	A,C,D	A,B,C	A,C,D	A,B,C	A,B	B	A
SECTION-I (ii)	Q.	9	10	11	12				
	A.	C	A	D	B				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	1.00	2.00	1.94	11.50	1.00	0.65		

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

1. Ans (A,B,C,D)



$$100 \times x_0 = 300 \times \frac{1}{100} \Rightarrow x_0 = \frac{3}{100} \text{ m}$$

$$k_{eq} = k_1 + k_2 = 400 \text{ N/m}$$

$$\omega = \sqrt{\frac{k}{m}} = 20 \text{ \& } T = \frac{2\pi}{\omega} = \frac{\pi}{10} = 0.314 \text{ sec}$$

$$\& V_{max} = A\omega$$

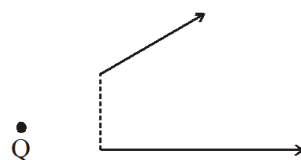
$$\text{if } A = \frac{1}{100} \text{ m}$$

$$\Rightarrow v_{max} = 0.2 \text{ m/s}$$

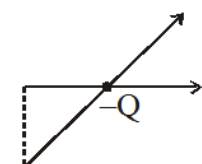
$$\text{if } A = \frac{3}{100} \text{ m}$$

$$\Rightarrow v_{max} = 0.3 \text{ m/s}$$

2. Ans (A,C)



(a) Larger the distance shorter the field vector.



(b) A negative charge at point of intersection.

3. Ans (A,B,D)

$$\mu_0 (I_1 + I_2) = 2\mu_0$$

$$\mu_0 (I_1 + I_3) = 4\mu_0$$

$$\mu_0 (I_1 + I_2 + I_3) = \mu_0$$

$$\Rightarrow I_1 = 5A \text{ \& } I_3 = -1A$$

$$I_2 = -3A$$

7. Ans (A,C,D)

$$(A) f = \frac{1}{2\ell} \sqrt{\frac{T}{\mu}} \alpha \sqrt{T}$$

$$\frac{f_2}{f_1} = \sqrt{\frac{T_2}{T_1}} = \sqrt{\frac{1.21T}{T}} = 1.1$$

$$f_2 = 10\% \text{ more than } f_1$$

$$\Delta f = (0.1) f_1 = 15 \text{ Hz}$$

$$f_1 = 150 \text{ Hz}$$

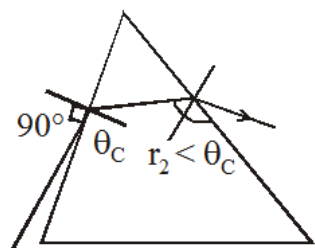
$$(C) v = \sqrt{\frac{T}{\mu}} \alpha \sqrt{T}$$

$$\frac{v_2}{v_1} = \sqrt{\frac{T_2}{T_1}} = \sqrt{1.21} = 1.1$$

$$v_2 = 10\% \text{ more than } v_1$$

$$(D) \lambda = 2L \text{ remains unchanged.}$$

8. Ans (A,B,C)



$$\mu \sin \theta_C = 1$$

$$\theta_C = 53^\circ$$

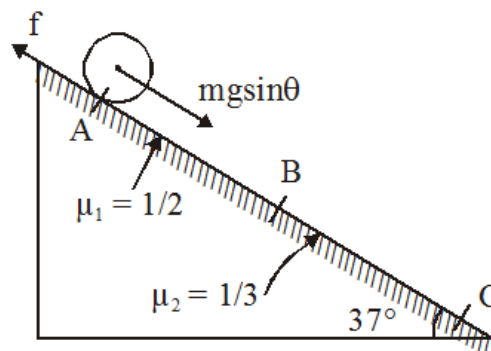
$$A = r_1 + r_2 < 2\theta_C$$

$$A < 106^\circ$$

PART-1 : PHYSICS

SECTION-II

1. Ans (0.41)



For pure rolling

$$mg \sin \theta - f = ma \dots (1)$$

$$\frac{mR^2}{2} \times \alpha = f \times R \dots (2)$$

$$R\alpha = a \dots (3)$$

From (1), (2) & (3)

$$\Rightarrow \frac{2f}{m} = \frac{mg \sin \theta - f}{m} \Rightarrow f = \frac{mg \sin \theta}{3}$$

$$\& f \leq \mu N \frac{mg \sin \theta}{3} \leq \mu mg \cos \theta$$

$$\Rightarrow \mu \geq \frac{\tan \theta}{3} \Rightarrow \mu \geq \frac{1}{4} \text{ [For pure rolling]}$$

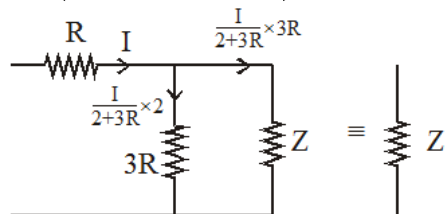
$\Rightarrow$ disc will pure roll all the time

$$\text{Now } AB = \frac{1}{2} at_{AB}^2 \text{ \& } t_{BC} = t_{AC} - t_{AB}$$

$$AC = 2AB = \frac{1}{2} at_{AC}^2$$

$$\Rightarrow \frac{t_{BC}}{t_{AB}} = \frac{t_{AC}}{t_{AB}} - 1 = \sqrt{2} - 1 = 0.41$$

2. Ans (111.00 to 111.90)

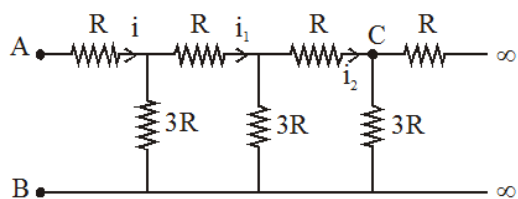


$$\begin{aligned}\text{Now } R + \frac{3RZ}{3R+2} &= Z \\ 3R^2 + RZ + 3RZ &= 3RZ + Z^2 \\ Z^2 + RZ + 3RZ &= 3RZ + Z^2 \\ Z^2 - RZ - 3R^2 &= 0\end{aligned}$$

$$\Rightarrow Z = \frac{R + \sqrt{R^2 + 12R^2}}{2}$$

$$Z = R \left(\frac{1 + \sqrt{13}}{2} \right) = 2.3R$$

$$\Rightarrow \frac{R}{Z} = \frac{1}{2.3}$$



$$i = \frac{V_{AB}}{Z} = \frac{\Delta V}{Z}$$

$$i_1 = \frac{i}{Z + 3R} \times 3R = \frac{\Delta V}{Z} \times \left(\frac{3R}{Z + 3R} \right)$$

$$i_2 = \frac{i_1}{Z + 3R} \times 3R = \frac{\Delta V}{Z} \left(\frac{3R}{Z + 3R} \right)^2$$

$$\Delta V_{AC} = \Delta C - iR - i_1 R - i_2 R$$

$$\Delta V_{AC} = \Delta V$$

$$\left[1 - \frac{R}{2} - \frac{R}{2} \left(\frac{3R}{Z + 3R} \right) - \frac{R}{2} \left(\frac{3R}{Z + 3R} \right)^2 \right]$$

$$\therefore \left(\frac{3R}{Z + 3R} \right) = \frac{3}{\frac{Z}{R} + 3} = \frac{3}{5.3}$$

Now

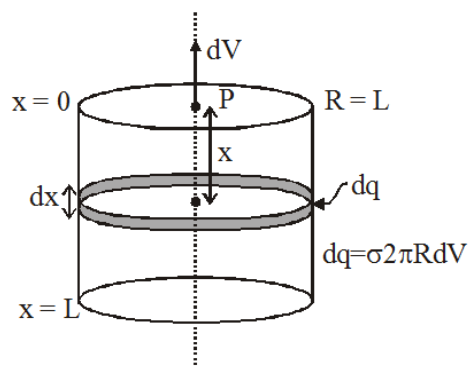
$$20 = \Delta V \left[1 - \frac{1}{2.3} \left\{ 1 + \frac{3}{5.3} + \left(\frac{3}{5.3} \right)^2 \right\} \right]$$

$$20 = \Delta V \left[1 - \frac{1}{2.3} \{ 1 + 0.566 + 0.32 \} \right]$$

$$20 = \Delta V \left[1 - \frac{1.88}{2.3} \right] = \Delta V \times 0.18$$

$$\Delta V = 111.11$$

3. Ans (2.00)



$$dV = \frac{1}{4\pi\epsilon_0} dq$$

$$\Rightarrow \int_0^V dV = \int_0^L \frac{\sigma 2\pi L dx}{4\pi\epsilon_0 (x^2 + L^2)^{\frac{1}{2}}}$$

$$\Rightarrow V = \frac{\sigma L}{2\epsilon_0} \ln(1 + \sqrt{2})$$

4. Ans (2.50 to 2.59)

A bright line for the green light requires

$$d\sin\theta = d\tan\theta = m_1\lambda_1$$

$$d\frac{y}{L} = m_1 \lambda_1$$

Similarly, a blue interference maximum requires

$$d\frac{y}{L} = m_2\lambda_2$$

for integers m_1 and m_2 . Thus,

$$m_1(540 \text{ nm}) = m_2(450 \text{ nm})$$

$$\frac{m_2}{m_1} = \frac{540 \text{ nm}}{450 \text{ nm}} = \frac{6}{5}$$

and smallest integers satisfying the equation are

$m_1 = 5$ and $m_2 = 6$.

Then for both,

$$d\frac{y}{L} = 2700 \text{ nm}$$

which gives

$$y = (2.700 \text{ nm}) \frac{L}{d} = (2.7 \mu\text{m}) \left(\frac{1.4 \text{ m}}{150 \mu\text{m}} \right) = 2.52 \text{ cm}$$

5. Ans (242.00 to 242.90)

$$\frac{hc}{\lambda_1} = 13.6\text{eV} \left[1 - \frac{1}{2^2} \right]$$

$$\lambda_1 = 121.76 \Rightarrow \lambda_2 - \lambda_1 = 243.53$$

$$\frac{hc}{\lambda_2} = 13.6\text{eV} \left[\frac{1}{2^2} - \frac{1}{\infty^2} \right]$$

$$\lambda_2 = 365.29$$

6. Ans (0.71 to 0.73)

$$\frac{hc}{\lambda} = (13.6\text{eV}) (Z-1)^2 \left[1 - \frac{1}{(2)^2} \right]$$

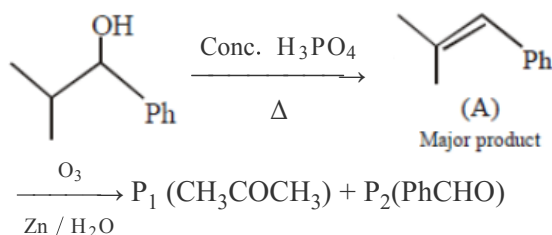
$$\frac{12400}{\lambda} = 13.6 \times (41)^2 \times \frac{3}{4}$$

$$\Rightarrow \lambda = 0.72 \text{ \AA}$$

PART-2 : CHEMISTRY

SECTION-I (i)

2. Ans (A,C,D)



Both compounds P_1 and P_2 reacts with

NaHSO_3 .

P_2 reduces tollens reagent but can not reduce

Fehling's solution.

Only P_1 can undergo iodoform test.

Compound P_2 can undergo positive 2,4-DNP

test as well as Cannizzaro reaction.

5. Ans (A,C,D)

$$\frac{t_{50\%}}{t_{75\%}} = \frac{\log_{10} \left(\frac{100}{100-50} \right)}{\log_{10} \left(\frac{100}{100-75} \right)}$$

$$= \frac{\log_{10} 2}{\log_{10} 4} = \frac{\log_{10} 2}{2 \cdot \log_{10} 2}$$

$$\frac{t_{50\%}}{t_{75\%}} = \frac{1}{2} = D \quad t_{75\%} = 2t_{50\%}$$

$$K = A \cdot e^{-E_a/RT}$$

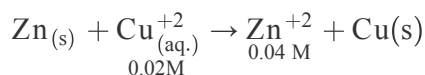
K increases exponentially with increase in temperature.

PART-2 : CHEMISTRY

SECTION-II

2. Ans (109.00)

Galvanic cell:



$$\text{Nernst equation} = E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.059}{2} \log \frac{[\text{Zn}^{+2}]}{[\text{Cu}^{+2}]}$$

$$\Rightarrow E_{\text{cell}} = \left[E_{\text{cell}}^{\circ} - E_{\text{Zn}^{+2}/\text{Zn}}^{\circ} \right] - \frac{0.059}{2} \log \frac{0.04}{0.02}$$

$$\Rightarrow E_{\text{cell}} = [0.34 - (-0.76)] - \frac{0.059}{2} \log^2$$

$$\Rightarrow E_{\text{cell}} = 1 - 1 - \frac{0.059}{2} \times 0.3010$$

$$= 1.0911 = 109.11 \times 10^{-2}$$

$$= 109$$

3. Ans (33.33)

$$5.58 = \frac{1.86 \times 12.4 \times 1000}{62 \times W}$$

$$W = \frac{12.4 \times 1000 \times 1.86}{5.58 \times 62} = \frac{200}{3}$$

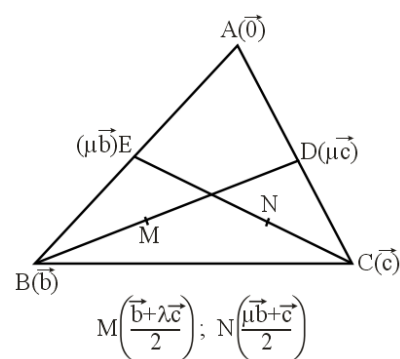
$$\text{Ice separated} = 100 - \frac{200}{3} = \frac{100}{3} = 33.33\text{g}$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,B,C)

$$\Delta = \frac{1}{2} |\vec{b} \times \vec{c}|;$$



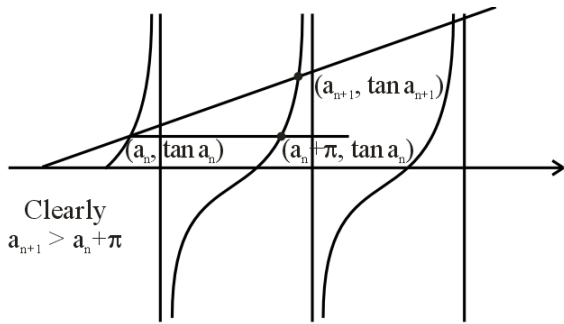
$$\Delta_1 = \frac{(1 - \mu\lambda) |\vec{b} \times \vec{c}|}{8}$$

$$\Delta_2 = \frac{1}{2} \mu\lambda |\vec{b} \times \vec{c}|$$

$$\Delta_3 = \frac{1}{2} (1 - \mu\lambda) |\vec{b} \times \vec{c}|$$

$$\Delta_4 = \frac{(1 - \mu\lambda) |\vec{b} \times \vec{c}|}{8}$$

2. Ans (A,C,D)



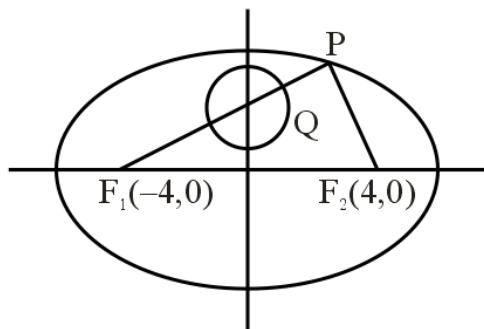
$$\frac{dy}{dx} = \cos x \Big|_{x=a_n} = \frac{\sin a_n}{a_n + \frac{\pi}{2}}$$

$$a_n + \frac{\pi}{2} = \tan a_n$$

Taking intersection of $y = \tan x$

$$\text{and } y = x + \frac{\pi}{2}$$

3. Ans (A,B,C)



$$PQ + PF_2 = 2a - QF_1$$

$$(PQ + PF_2)_{\max} = 12 - (QF_1)_{\min} = 12 - (4) = 8$$

$$(PQ + PF_2)_{\min} = 12 - (QF_1)_{\max} = 12 - 6 = 6$$

For option C, PF_1 is tangent to circle

$$\text{So } QF_1 = \sqrt{24} \text{ so } PQ + PF_2 = 12 - \sqrt{24}$$

4. Ans (A,C,D)

$$\begin{aligned} I_{n+1} + I_{n-1} &= \int_0^\pi \left[\frac{1 - \cos((n+1)x)}{1 - \cos x} + \frac{1 - \cos((n-1)x)}{1 - \cos x} \right] dx \\ &= \int_0^\pi \frac{2 - 2(\cos nx)(\cos x)}{1 - \cos x} dx \\ &= 2 \int_0^\pi \frac{1 - (\cos nx)(\cos x)}{1 - \cos x} dx \\ &= 2 \int_0^\pi \frac{1 - \cos nx + \cos nx(1 - \cos x)}{1 - \cos x} dx \\ &= 2I_n + 2 \underbrace{\int_0^\pi \cos nx dx}_{=0} \end{aligned}$$

$$\frac{I_{n+1} + I_{n-1}}{2} = I_n ; \forall n \geq 1$$

$$\text{Now } I_0 = 0 ; I_1 = \int_0^\pi \frac{1 - \cos x}{1 - \cos x} dx = \pi$$

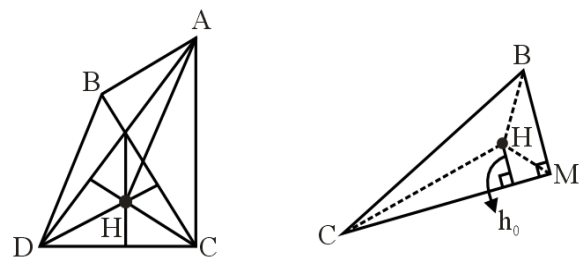
$$\text{Hence } I_n = n\pi ;$$

$$\therefore I_0, I_1, I_2, \dots \text{ are in A.P.}$$

Hence A, C, D is correct.

5. Ans (A,B,C)

Let $\perp$ distance from H to BC, BD and CD be



$$p, 2p, 3p$$

$$\frac{1}{2} \times 12(p + 2p + 3p) = \frac{1}{2} 12^2 \sin 60^\circ$$

$$p = \sqrt{3}$$

$$\begin{aligned} &\frac{1}{2} \times 6 \times 2\sqrt{3} + \frac{1}{2} \times 12\sqrt{3} + \frac{1}{2} \times h_0 6\sqrt{3} \\ &= \frac{1}{2} \left(\frac{1}{2} \times 12^2 \sin 60^\circ \right) \end{aligned}$$

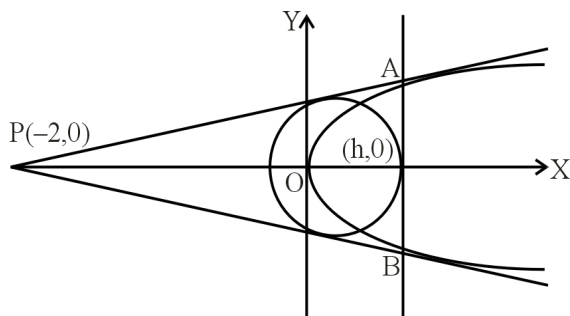
$$\therefore h_0 = 2. \text{ So } \perp \text{ distance of A from CM}$$

$$= \sqrt{2^2 + 3^2} = \sqrt{13}$$

6. **Ans (A, B)**

Point P clearly lies on the directrix of $y^2 = 8x$. Thus, slopes of PA and PB are 1 and -1 respectively.

Equation of PA : $y = x + 2$,



Equation of PB : $y = -x - 2$,

Equation of AB : $x - 2$

Let the centre of the circle be $(h, 0)$ and radius be r .

$$\Rightarrow \frac{|h+2|}{\sqrt{2}} = \frac{|h-2|}{1} = r$$

$$\Rightarrow h^2 + 4 + 4h = 2(h^2 + 4 - 4h)$$

$$\Rightarrow h^2 - 12h + 4 = 0$$

$$\therefore h = \frac{12 \pm 8\sqrt{2}}{2} = 6 \pm 4\sqrt{2}$$

$$\Rightarrow \text{radius} = |h-2| = 4(\sqrt{2}-1), 4(\sqrt{2}+1).$$

7. **Ans (B)**

Equation of any plane through the line is a

$$(x-1) + b(y+1) + c(z-3) = 0$$

$$\text{where } 2a - b + 4c = 0 \quad \dots(i)$$

The plane $x + 2y + z = 6$ perpendicular

$$\text{to } a(x-1) + b(y+1) + c(z-3) = 0$$

$$\therefore a + 2b + c = 0 \quad \dots(ii)$$

$$\text{Solving (i) \& (ii) we get } \frac{a}{-9} = \frac{b}{2} = \frac{c}{5}$$

$$\text{equation of the plane is } 9x - 2y - 5z + 4 = 0$$

Hence the equation of projection is

$$9x - 2y - 5z + 4 = 0$$

8. **Ans (A)**

$$A^3 = B^3 \quad \dots(i)$$

$$A^2B = B^2A \quad \dots(ii)$$

Subtracting (ii) by (i)

$$(A^2 + B^2)(A - B) = 0$$

$$\therefore |A - B| \neq 0, A^2 + B^2 = 0$$

PART-3 : MATHEMATICS

SECTION-I (ii)

9. **Ans (C)**

Hint : (I) Find domain

$$(II) f(x) = 3 - 2\{x\}$$

$$(III) f(-x) = -f(x) \text{ and } f(x) = \begin{cases} \frac{x}{1+x} & \text{if } x \geq 0 \\ \frac{x}{1-x} & \text{if } x < 0 \end{cases}$$

$$(IV) f(-x) = f(x) \text{ and } x + \sqrt{1+x^2} \geq 1 \quad \forall x \geq 0$$

Solution :

$$(I) f(x) = \tan x - \sin x \quad \forall x \in \left[2n\pi, 2n\pi + \frac{\pi}{2}\right)$$

If $f(x)$ exists the $f(-x)$ does not exist. So $f(x)$ is neither even nor odd.

$$\text{Also } f(x) \geq 0 \quad \forall x \in D_f \text{ (not surjective)}$$

$$\text{and } f(2\pi + x) = f(x) \text{ (periodic and many one)}$$

$$(II) f(x) = 2[x] - 2x + 3 = 3 - 2\{x\}$$

$$\text{Range of } f(x) \text{ is } (1, 3] \text{ (not surjective)}$$

$$f(x) \text{ is periodic and not injective.}$$

$$(III) f(-x) = -f(x) \quad \text{(odd function)}$$

$$f(x) = \begin{cases} 1 - \frac{1}{1+x} & \text{if } x \geq 0 \\ -1 + \frac{1}{1-x} & \text{if } x < 0 \end{cases}$$

$$\text{Range of } f(x) \text{ is } (-1, 1) \text{ (surjective)}$$

$$\text{and } f'(x) > 0, f(x) \text{ is strictly increasing (one-one)}$$

$$(IV) f(-x) = \left(\ln(\sqrt{1+x^2} - x)\right)^2$$

$$\Rightarrow f(-x) = \left(\ln\left(\frac{1}{\sqrt{1+x^2} + x}\right)\right)^2$$

$$\Rightarrow f(-x) = \left(\ln(x + \sqrt{1+x^2})\right)^2 = f(x)$$

$$\text{(even function and many one)}$$

$$\text{Also } x + \sqrt{1+x^2} \geq 1 \quad \forall x \geq 0$$

$$\Rightarrow \ln(x + \sqrt{1+x^2}) \geq 0 \quad \forall x \geq 0$$

$$\text{Range of } f(x) \text{ is } [0, \infty) \text{ (surjective)}$$

10. Ans (A)

Hint : (I) Find domain

(II) $f(x) = 3 - 2\{x\}$

(III) $f(-x) = -f(x)$ and $f(x) = \begin{cases} \frac{x}{1+x} & \text{if } x \geq 0 \\ \frac{x}{1-x} & \text{if } x < 0 \end{cases}$

(IV) $f(-x) = f(x)$ and $x + \sqrt{1+x^2} \geq 1 \forall x \geq 0$

Solution :

(I) $f(x) = \tan x - \sin x \forall x \in \left[2n\pi, 2n\pi + \frac{\pi}{2}\right)$

If $f(x)$ exists the $f(-x)$ does not exist. So $f(x)$ is neither even nor odd.

Also $f(x) \geq 0 \forall x \in D_f$ (not surjective)

and $f(2\pi + x) = f(x)$ (periodic and many one)

(II) $f(x) = 2[x] - 2x + 3 = 3 - 2\{x\}$

Range of $f(x)$ is $(1, 3]$ (not surjective)

$f(x)$ is periodic and not injective.

(III) $f(-x) = -f(x)$ (odd function)

$f(x) = \begin{cases} 1 - \frac{1}{1+x} & \text{if } x \geq 0 \\ -1 + \frac{1}{1-x} & \text{if } x < 0 \end{cases}$

Range of $f(x)$ is $(-1, 1)$ (surjective)

and $f'(x) > 0$, $f(x)$ is strictly increasing (one-one)

(IV) $f(-x) = \left(\ln\left(\sqrt{1+x^2} - x\right)\right)^2$

$\Rightarrow f(-x) = \left(\ln\left(\frac{1}{\sqrt{1+x^2} + x}\right)\right)^2$

$\Rightarrow f(-x) = \left(\ln\left(x + \sqrt{1+x^2}\right)\right)^2 = f(x)$

(even function and many one)

Also $x + \sqrt{1+x^2} \geq 1 \forall x \geq 0$

$\Rightarrow \ln\left(x + \sqrt{1+x^2}\right) \geq 0 \forall x \geq 0$

Range of $f(x)$ is $[0, \infty)$ (surjective)

11. Ans (D)

$\Delta = \begin{vmatrix} a^2 & a^2 - (b-c)^2 & bc \\ b^2 & b^2 - (c-a)^2 & ca \\ c^2 & c^2 - (a-b)^2 & ab \end{vmatrix} \quad C_2 \rightarrow C_2 - C_1$

$= - \begin{vmatrix} a^2 & (b-c)^2 & bc \\ b^2 & (c-a)^2 & ca \\ c^2 & (a-b)^2 & ab \end{vmatrix} \quad C_2 \rightarrow C_2 + 2C_3$

$= - \begin{vmatrix} a^2 & b^2 + c^2 & bc \\ b^2 & c^2 + a^2 & ca \\ c^2 & a^2 + b^2 & ab \end{vmatrix}$

$= -(a^2 + b^2 + c^2) \begin{vmatrix} 1 & b^2 + c^2 & bc \\ 1 & c^2 + a^2 & ca \\ 1 & a^2 + b^2 & ab \end{vmatrix}$

$C_1 \rightarrow C_1 + C_2$

$= -(a^2 + b^2 + c^2) \begin{vmatrix} 0 & (b-a)(b+a) & c(b-a) \\ 0 & (c-b)(c+b) & a(c-b) \\ 1 & a^2 + b^2 & ab \end{vmatrix}$

$R_1 \rightarrow R_1 - R_2, R_2 \rightarrow R_2 - R_3$

$= -(a^2 + b^2 + c^2)(b-a)(c-b)(a-c)(a+b+c)$
 $= (a-b)(b-c)(c-a)(a+b+c)(a^2 + b^2 + c^2)$

(I) If $a + b + c = 0 \Rightarrow \Delta = 0$

Given system is homogeneous system, therefore infinite number of solutions will exist

(II) If $a = b = c = 0$, then all three equation will be $0.x + 0.y + 0.z = 0$

therefore any (x, y, z) will satisfy the given set of equations where $x, y, z \in \mathbb{R}$.

(III) $a = b = c, a + b + c \neq 0$

then all three equations will come in form $x + y + z = 0$

solution of equation are $(\lambda_1, \lambda_2, -\lambda_1 - \lambda_2)$

(IV) $\Delta \neq 0$

$\Rightarrow$ only $(0, 0, 0)$ is the solution.

$\Rightarrow$ unique solution.

12. Ans (B)

$$\Delta = \begin{vmatrix} a^2 & a^2 - (b-c)^2 & bc \\ b^2 & b^2 - (c-a)^2 & ca \\ c^2 & c^2 - (a-b)^2 & ab \end{vmatrix} \quad C_2 \rightarrow C_2 - C_1$$

$$= - \begin{vmatrix} a^2 & (b-c)^2 & bc \\ b^2 & (c-a)^2 & ca \\ c^2 & (a-b)^2 & ab \end{vmatrix} \quad C_2 \rightarrow C_2 + 2C_3$$

$$= - \begin{vmatrix} a^2 & b^2 + c^2 & bc \\ b^2 & c^2 + a^2 & ca \\ c^2 & a^2 + b^2 & ab \end{vmatrix}$$

$$= -(a^2 + b^2 + c^2) \begin{vmatrix} 1 & b^2 + c^2 & bc \\ 1 & c^2 + a^2 & ca \\ 1 & a^2 + b^2 & ab \end{vmatrix}$$

$$C_1 \rightarrow C_1 + C_2$$

$$= -(a^2 + b^2 + c^2) \begin{vmatrix} 0 & (b-a)(b+a) & c(b-a) \\ 0 & (c-b)(c+b) & a(c-b) \\ 1 & a^2 + b^2 & ab \end{vmatrix}$$

$$R_1 \rightarrow R_1 - R_2, R_2 \rightarrow R_2 - R_3$$

$$= -(a^2 + b^2 + c^2)(b-a)(c-b)(a-c)(a+b+c)$$

$$= (a-b)(b-c)(c-a)(a+b+c)(a^2 + b^2 + c^2)$$

(I) If $a + b + c = 0 \Rightarrow \Delta = 0$

Given system is homogeneous system,
therefore infinite number of solutions will exist
(II) If $a = b = c = 0$, then all three equation will
be $0.x + 0.y + 0.z = 0$

therefore any (x, y, z) will satisfy the given
set of equations where $x, y, z \in \mathbb{R}$.

(III) $a = b = c, a + b + c \neq 0$

then all three equations will come in form x
 $+ y + z = 0$

solution of equation are $(\lambda_1, \lambda_2, -\lambda_1 - \lambda_2)$

(IV) $\Delta \neq 0$

$\Rightarrow$ only $(0, 0, 0)$ is the solution.

$\Rightarrow$ unique solution.

PART-3 : MATHEMATICS

SECTION-II

1. Ans (1.00)

Hint : $\alpha, \beta \in (0, 1)$

Solution :

Let $y = x^x$

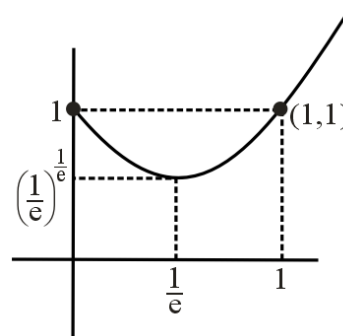
$$\frac{dy}{dx} = x^x(1 + \ln x)$$

So $f(x)$ is decreasing in $(0, \frac{1}{e})$ and

increasing in $(\frac{1}{e}, \infty)$

$$\text{Also } \lim_{x \rightarrow 0^+} x^x = e^{\lim_{x \rightarrow 0^+} x \ln x} = 1$$

So we can draw the graph $y = f(x)$



If $f(x) = k$ has two distinct solutions α, β , then

$$k \in \left(\left(\frac{1}{e} \right)^{1/e}, 1 \right)$$

Also $\alpha, \beta \in (0, 1)$

Then $|\alpha - \beta| < 1$

2. Ans (2.00)

$$\int_1^e \frac{\frac{1 - \ln x}{x^2}}{\sqrt{1 + \left(\frac{\ln x}{x} \right)^2}} dx \quad \text{Let } \frac{\ln x}{x} = t$$

$$\frac{1 - \ln x}{x^2} dx = dt$$

$$= \int_0^{1/e} \frac{dt}{\sqrt{1 + t^2}}$$

$$= \ln \left(t + \sqrt{1 + t^2} \right) \Big|_0^{1/e}$$

$$= \ln \left(1 + \sqrt{1 + e^2} \right)$$

3. Ans (1.94)

Hint: Use binomial distribution

Solution: Probability that a question with only one option correct is correct = $\frac{1}{4}$.

Probability that a question with one or more option correct is correct = $\frac{1}{15}$

If a student is getting 4 marks then there will be two cases only.

(i) 2 single option correct and one more than one option correct

(ii) No single option correct and two more than one option correct

$$\text{required probability} = {}^3C_2 \cdot \left(\frac{1}{4}\right)^2 \cdot \frac{3}{4} \cdot {}^2C_1 \cdot \left(\frac{1}{15}\right)^1 \\ + {}^3C_0 \cdot \left(\frac{1}{4}\right)^0 \cdot \left(\frac{3}{4}\right)^3 \cdot {}^2C_2 \cdot \left(\frac{1}{15}\right)^2 \cdot \left(\frac{14}{15}\right)^0 \\ = \frac{28}{1600} + \frac{3}{1600} = \frac{31}{1600}$$

$$\text{So, } k = \frac{31}{16}$$

4. Ans (11.50)

$$\sqrt{x} = t$$

$$y = t^3 + \frac{1}{8t^3} - \frac{17}{4} \left(t^2 + \frac{1}{4t^2} \right)$$

$$y = \left(t + \frac{1}{2t} \right)^3 - \frac{3}{2} \left(t + \frac{1}{2t} \right) - \frac{17}{4} \left(\left(t + \frac{1}{2t} \right)^2 - 1 \right)$$

$$y = u^3 - \frac{3}{2}(u) - \frac{17}{4}(u^2 - 1) \quad (\text{where}$$

$$u = t + \frac{1}{2t}) \text{ which has minimum at } u = 3$$

$$y(3) = 27 - \frac{9}{2} - \frac{17}{4}(8) = -\frac{23}{2}$$

5. Ans (1.00)

$$q = \sum_{r=1}^{\infty} \frac{1}{2^r} \left(1 - \frac{1}{2^r} \right)^{2021} \\ = \sum_{r=1}^{\infty} \sum_{k=0}^{2021} (-1)^k \frac{{}^{2021}C_k}{2^{(k+1)r}} \\ = \sum_{k=0}^{2021} (-1)^k {}^{2021}C_k \sum_{r=1}^{\infty} \frac{1}{2^{(k+1)r}} \\ = \sum_{k=0}^{2021} (-1)^k \frac{{}^{2021}C_k}{(2^{k+1} - 1)} \\ = 1 - \sum_{k=1}^{2021} \frac{(-1)^{k-1} {}^{2021}C_k}{2^{k+1} - 1}$$

$$q = 1 - p$$

6. Ans (0.65)

$$\alpha, \beta = \frac{a + 5 \pm \sqrt{(a + 5)^2 - 8a}}{2}$$

for α, β to be rational

Hence $(a + 5)^2 - 8a = \text{perfect square of some integer}$

$$a^2 + 2a + 25 = N^2$$

$$24 = N^2 - (a + 1)^2$$

$$C-I : N^2 = 49$$

$$a + 1 = \pm 5 \begin{cases} a=4 \\ a=-6 \end{cases}$$

$$C-II : N^2 = 25$$

$$a + 1 = \pm 1 \begin{cases} a=0 \\ a=-2 \end{cases}$$

Hence for $a = 4$, $(\alpha, \beta) \equiv (1, 8)$

$$a = -6, (\alpha, \beta) \equiv (-4, 3)$$

$$a = 0; (\alpha, \beta) \equiv (0, 5)$$

$$a = -2; (\alpha, \beta) \equiv (-1, 4)$$

$$(\alpha^2 + \beta^2)_{\max} = 65$$